Global & Indian Telehealth Replica Recon Report

Deep Comparative Analysis: India, United States, United Kingdom, and European Union

Document Reference: KGS-RECON-V2.0 • Prepared for Kaizentrix Global Solutions

DEEP GLOBAL & INDIAN TELEHEALTH REPLICA RECON REPORT

Comprehensive Reverse-Engineering Analysis of Leading Digital Health Platforms

India (Practo, Apollo 24|7, eSanjeevani), United States (Teladoc, Amwell), United Kingdom (Livi, Doctor Care Anywhere), European Union (Doctolib France & Germany)

Scope: End-to-End Application Flows, Screen Inventories, Component Models, Technical Architectures, and Engineering Blueprint for a Globally Extensible Telehealth System
Prepared for: Kaizentrix Global Solutions
Date: October 2026 (Updated International Baseline)

Global & Indian Telehealth Ecosystem Reconnaissance Map

Comprehensive Reverse-Engineering Blueprint: Indian & International Digital Health Platforms

Scope: End-to-End Telehealth, Doctor Discovery, EHR/HIS Integration, E-Prescribing, Insurance Clearinghouses, Multi-Party Video, Clinical Governance, and Sovereign Digital Health Infrastructures
Platform Coverage: Responsive Web (Desktop/Mobile), Native iOS, Native Android, Embedded EHR iframes (SMART on FHIR)
Target Reference Ecosystems:
* **India (Launch Market):** Practo (Commercial Marketplace/SaaS), Apollo 24|7 (Hospital Omnichannel), eSanjeevani 2.0 (Public Infrastructure / C-DAC / MoHFW).
* **United States (Enterprise / Payer Benchmark):** Teladoc Health (Employer/Health Plan Primary Care & Chronic Care), Amwell Converge™ (Hospital EHR-Integrated Hybrid Care & Multi-Party Video).
* **United Kingdom (NHS Primary Care Benchmark):** Livi (NHS GP Connect & Electronic Prescription Service), Doctor Care Anywhere (Private Medical Insurance & Diagnostic Referral Pathways).
* **European Union (Sovereign Social Insurance Benchmark):** Doctolib France (Ameli Tiers-Payant, Mon Espace Santé & HDS Certification), Doctolib Germany (GKV/PKV, Gematik Telematikinfrastruktur, e-Rezept & ePA).
Date: October 2026 (Updated Baseline)

1. Sources & Methodology

#	Platform	Jurisdiction	Source Type	URL / Reference	Notes & Architectural Coverage
1	**Practo**	India	Patient Web & Mobile	`https://www.practo.com/`	Doctor search, verified profiles, in-clinic booking, Practo Consult instant video, reviews.
2	**Practo Ray**	India	Clinic Practice SaaS	`https://www.practo.com/ray`	Multi-doctor calendar, EMR charting, walk-in management, billing, patient records.
3	**Apollo 24|7**	India	Enterprise Hospital Portal	`https://www.apollo247.com/`	Hospital doctor consults, Express 15-min consult, Clinical Intelligence Engine (CIE/MedLM).
4	**eSanjeevani 2.0**	India	National Sovereign OPD	`https://esanjeevani.mohfw.gov.in/`	Citizen-to-Doctor OPD token queue and AB-HWC Hub-and-Spoke rural assisted telemedicine.
5	**ABDM Ecosystem**	India	National Health Authority	`https://abdm.gov.in/`	Ayushman Bharat Health Account (ABHA M1), Health Information Provider/User (M2/M3 FHIR).
6	**Teladoc Health**	United States	Enterprise Virtual Care	`https://www.teladoc.com/`	Employer/health plan eligibility (270/271), Primary360, Livongo chronic care, Surescripts e-Rx.
7	**Amwell Converge™**	United States	Hospital Telehealth EHR	`https://business.amwell.com/`	SMART on FHIR Epic/Cerner embedded launch, multi-party video (interpreter, family), multi-state licensure.
8	**Livi (Kry)**	United Kingdom	NHS Primary Care Partner	`https://www.livi.co.uk/`	NHS login SSO, GP Connect record write-back (EMIS Web / SystmOne), Electronic Prescription Service (EPS).
9	**Doctor Care Anywhere**	United Kingdom	Private Medical Insurer	`https://doctorcareanywhere.com/`	AXA Health / Bupa integration, private specialist referrals, diagnostic pathway management.
10	**Doctolib France**	France / EU	National Health Portal	`https://www.doctolib.fr/`	Carte Vitale / Ameli billing, Mon Espace Santé (DMP), MSSanté secure messaging, HDS cloud host certified.
11	**Doctolib Germany**	Germany / EU	Telematics Health Portal	`https://www.doctolib.de/`	GKV/PKV statutory insurance, Gematik Telematikinfrastruktur, e-Rezept QR code, ePA patient file.

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2. Core Operational Loops by Jurisdiction

2.1 India Market Loops

- **Practo Core Loop:** Patient discovers clinician via geo-specialty search & reviews -> Books slot or instant consult -> Doctor charts encounter notes and issues digital e-prescription -> Patient receives one-tap fulfillment for pharmacy and home lab testing.
- **Apollo 24|7 Core Loop:** Patient completes AI symptom triage (CIE) or selects hospital consultant -> Joins scheduled or 15-minute express queue -> Attending hospital physician issues e-prescription integrated with Apollo Hospital MedSys HIS.
- **eSanjeevani Core Loop:** Rural citizen visits Ayushman Bharat Health & Wellness Centre (Spoke) -> Frontline CHO logs vitals -> Cloud router queues case to Medical College Hub -> Specialist conducts WebRTC consult and issues digitally signed Rx -> CHO dispenses generic drugs on-site at zero cost.

2.2 United States Market Loops

- **Teladoc Health Core Loop:** Insured employee signs in -> Platform runs real-time 270/271 EDI eligibility check with health plan (e.g., Aetna/BCBS) -> Patient selects virtual primary care provider or acute queue -> Provider reviews past records and conducts HIPAA-compliant video session -> Provider issues electronic prescription directly to patient's local retail pharmacy via Surescripts switch.
- **Amwell Converge™ Core Loop:** Clinician in hospital opens Epic Hyperspace schedule -> Clicks 'Start Video Visit' launching Amwell inside Epic via SMART on FHIR -> Patient joins from Epic MyChart mobile app -> Clinician brings in a certified Spanish Medical Interpreter and a remote family member into a 4-way call -> Consultation documentation and ICD-10/CPT codes automatically write back into Epic EHR repository.

2.3 United Kingdom Market Loops

- **Livi NHS Core Loop:** UK citizen logs in via NHS login -> System retrieves NHS Number and verifies GP practice registration via Personal Demographics Service (PDS) -> Citizen books free NHS GP video appointment -> GP accesses patient's historical medical summary via GP Connect API -> Consultation concludes with electronic prescription transmitted to patient's nominated pharmacy via NHS Electronic Prescription Service (EPS), and consultation notes automatically write back into the practice's EMIS Web or SystmOne clinical system.

2.4 European Union (France & Germany) Market Loops

- **Doctolib France Core Loop:** Patient searches specialist by sector/tarification -> System checks dynamic waiting list and offers earlier slot if another patient cancels (*Avancer mon rendez-vous*) -> Patient enters Carte Vitale health insurance info -> Secure WebRTC consultation conducted on HDS-certified cloud -> Doctor generates e-prescription with verification QR code -> Invoice submitted to French National Health Insurance (Ameli) via SESAM-Vitale tiers-payant; encounter summary automatically synced to citizen's *Mon Espace Santé* (DMP).
- **Doctolib Germany Core Loop:** Patient books statutory (GKV) or private (PKV) teleconsultation -> Doctor conducts consultation compliant with Gematik video guidelines -> Doctor digitally signs e-Rezept using German Health Professional Card (HBA) -> e-Rezept token dispatched to patient's Gematik app or redeemable at any German pharmacy via eGK card; notes synced to electronic patient dossier (ePA).

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3. Comprehensive Global Screen Inventory (S01 - S28)

ID	Screen Name	Route / Path	Reference Platforms	Primary Functional Goal	Observed UI States & Innovations
S01	Global Landing & Discovery	`/`	All Platforms	Value proposition, search, service entry	Geolocation prompt, category cards, insurance selector.
S02	Provider Directory & Filters	`/doctors`	All Platforms	Filter by specialty, fee, distance, insurance	Loading skeleton, map view, distance radius, verified badges.
S03	Provider Credentials & Profile	`/doctor/:id`	All Platforms	Council registration, degrees, patient stories	Verified badge, accepted insurances, clinic hours, bio.
S04	Slot Selection & Checkout	`/booking/:id`	Practo / Apollo / Doctolib	Date carousel, AM/PM chips, dependent picker	Slots open, slot locked, waitlist option if fully booked.
S05	Booking Confirmation Voucher	`/booking/success/:id`	All Platforms	Confirmed appointment pass & calendar sync	Success voucher, WhatsApp/SMS push, calendar invite (.ics).

ID	Screen Name	Route / Path	Reference Platforms	Primary Functional Goal	Observed UI States & Innovations
S06	Instant / Acute Teleconsult Queue	`/consult/instant`	Practo / Apollo / Teladoc	Dynamic matching with pooled on-call doctor	Symptom input, matching spinner, SLA counter (<15 mins).
S07	Video Teleconsultation Room	`/room/:id`	All Platforms	Encrypted live audio/video encounter	Device check, split grid, PiP, low-bandwidth audio-only.
S08	Digital Prescription Viewer	`/prescription/:id`	All Platforms	Legally signed prescription & advice PDF	Doctor digital signature, Schedule alert, pharmacy CTA.
S09	Digital Health Locker / EHR	`/vault`	Practo / Apollo / eSanj / Livi	Longitudinal medical history & records	Categorized records (Rx, labs, bills), ABHA/NHS sync.
S10	Doctor Practice Dashboard	`/doctor/dashboard`	Practo / Apollo / Doctolib	Daily schedule, queue, teleconsult toggle	Online/Offline toggle switch, daily stats, incoming alert.
S11	Clinical EMR Charting Console	`/emr/:patient`	Practo / Apollo / Amwell	Vitals, diagnosis, note builder, coding	ICD-10/SNOMED CT auto-suggest, clinical templates/macros.
S12	Front Desk / Triage Reception	`/clinic/reception`	Practo Ray / MedSys / AB-HWC	Walk-in check-in, vitals capture, cash desk	Live clinic queue, vitals entry modal, payment status.
S13	eSanjeevani Spoke CHO Console	`/hwc/dashboard`	eSanjeevani AB-HWC	Frontline health worker rural screening	Patient screening queue, state hub selector, call bridge.
S14	Admin Surveillance & Analytics	`/admin/analytics`	eSanjeevani / Doctolib	Operational volume, no-shows, wait times	Heatmap, conversion funnel, revenue ledger, error SLOs.
S15	Pre-Call Intake Questionnaire	`/intake/:apptId`	Teladoc / Livi / DCA	Capture symptoms, allergies, current drugs	Multi-step form, photo attachment, red-flag emergency alert.
S16	Real-Time Insurance Verification	`/insurance/verify`	Teladoc / Amwell / Doctolib	Real-time coverage & co-pay determination	Card photo OCR, 270/271 EDI query, Carte Vitale / GKV input.
S17	Multi-Party Video Room	`/room/multi/:id`	Amwell Converge / Teladoc	Connect patient, doctor, translator, family	4-way video grid, interpreter language picker, private chat.
S18	Embedded SMART on FHIR View	`EHR iframe / Tab`	Amwell Converge	Launch inside Epic / Cerner / EMIS Web	Context-aware patient header, one-click EHR note write-back.
S19	Dynamic Slot Pull-Forward	`/booking/pull-forward`	Doctolib ("Avancer RDV")	Offer earlier slot to patient when cancelled	Instant push notification, 10-minute accept countdown.
S20	Electronic Rx Routing Tracker	`/prescription/route`	Teladoc / Livi / Doctolib DE	Track digital Rx sent to pharmacy network	Surescripts / EPS / e-Rezept status (Sent, Ready for pickup).
S21	Dependent / Family Account View	`/account/family`	Practo / Teladoc / DCA	Manage child & elderly parent profiles	Dependent cards, relationship tag, pediatric dosing flags.
S22	Asynchronous Clinical Query	`/messages/:id`	Livi / Doctor Care Anywhere	7-day post-call follow-up chat with doctor	Secure messaging window, file attachment tray, SLA timer.

ID	Screen Name	Route / Path	Reference Platforms	Primary Functional Goal	Observed UI States & Innovations
S23	Diagnostic Lab Referral Slip	`/referral/lab/:id`	DCA / Practo / Apollo	Standardized pathology/radiology order slip	Clinical indications, fasting checklist, NABL/accredited lab.
S24	Multilingual Language Selector	Modal / Header	eSanjeevani / Doctolib	Switch platform language & localized UI	Indic languages (Hindi, Tamil), European (French, German).
S25	Emergency Redirection Takeover	Modal Overlay	All International Apps	Halts booking on acute emergency red-flags	High-contrast red banner, click-to-call 112/911/999, ER map.
S26	Provider Payout & Tax Ledger	`/finance/payouts`	Practo / Doctolib	Doctor earnings, commission split, TDS	Monthly payout summaries, TDS certificates, GST receipts.
S27	National Health ID SSO Auth	`/auth/national`	eSanjeevani / Livi / Doctolib	Login with ABHA (India), NHS login, Ameli	OTP verification, national health record consent gateway.
S28	Patient Verified Review Pad	`/reviews/new/:id`	Practo / Doctolib	Verified patient post-consultation feedback	1-5 star ratings (bedside manner, wait), moderation queue.

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4. Key International User Flows

Flow F04: US Health Plan / Employer Virtual Care with Surescripts e-Rx (Teladoc)

- **Goal:** Insured patient receives covered urgent care consultation with direct electronic prescription to nearest pharmacy.
- **Screens:** `S01 (Home)` -> `S16 (Insurance 270/271 Check)` -> `S15 (Intake & Pharmacy Picker)` -> `S06 (Urgent Queue)` -> `S07 (HIPAA WebRTC Call)` -> `S20 (Surescripts Dispatch)`.
- **Flow Details:** Patient selects employer benefit -> System validates zero copay via real-time clearinghouse -> Patient selects local CVS pharmacy -> Doctor reviews state licensure eligibility -> Video call conducted -> Doctor e-prescribes via Surescripts network -> Patient picks up medication 1 hour later.

Flow F05: UK NHS Integrated Teleconsultation via NHS login & GP Connect (Livi)

- **Goal:** UK citizen consults NHS GP remotely with automatic medical record write-back into GP surgery system.
- **Screens:** `S01` -> `S27 (NHS login SSO)` -> `S04 (NHS Slot Picker)` -> `S15 (Intake)` -> `S07 (WebRTC Room)` -> `S09 (GP Connect Sync)`.
- **Flow Details:** Patient authenticates using verified NHS login -> System retrieves NHS Number and links to registered surgery -> Video consultation conducted -> GP issues prescription via Electronic Prescription Service (EPS) -> Full consultation encounter bundle (diagnoses, medications, notes) writes back directly into surgery's EMIS Web or SystmOne record via GP Connect API.

Flow F06: European Sovereign Health Visit with Dynamic Slot Pull-Forward (Doctolib)

- **Goal:** European patient books specialist appointment, gets moved to an earlier slot automatically upon cancellation, and submits social insurance.
- **Screens:** `S01` -> `S02 (Specialist Search)` -> `S04 (Slot Picker with Waitlist Opt-in)` -> `S19 (Slot Pull-Forward Notification)` -> `S07 (HDS WebRTC Room)` -> `S08 (Signed e-Rezept / Ameli Sync)`.

- **Flow Details:** Patient books slot for 2 weeks out and toggles "Notify me if earlier slot opens" -> Another patient cancels 3 days later -> System sends instant push notification with 10-minute priority countdown -> Patient claims earlier slot -> Video call held on HDS-certified cloud -> Doctor generates e-Rezept / Carte Vitale reimbursement invoice.

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5. UI/UX Component Catalog (Extended Global Library)

Component Name	Category	Variants	Interactive States	Primary Screens
Multi-Party Video Viewport	Media Display	2-Way, 3-Way, 4-Way Grid, PiP	Connecting, Active HD, Audio-Only, Interpreter Active	`S07`, `S17`
Language Interpreter Picker	In-Call Tool	Floating Drawer, In-Call Pill	Available, Connecting Translator, Active Translation	`S17`
Insurance Eligibility Badge	Billing Widget	Active Covered, Copay Required, Ineligible	Verified Green, Pending Check, Verification Error	`S04`, `S16`
Dynamic Slot Pull-Forward Banner	Notification	Top Toast, Modal Alert	10-min Countdown, Accepted, Slot Expired	`S04`, `S19`
Camera Switcher Button	In-Call Tool	Front-Facing, Rear High-Res	Front Active, Rear Active, Flashlight Toggle	`S07`
In-Call Clinical Chat Drawer	In-Call Tool	Slide-Out Drawer, Overlay	Closed, New Message Badge, Open, Typing	`S07`, `S17`
National Health Identifier Badge	Identity Widget	ABHA (India), NHS (UK), Ameli (FR), eGK (DE)	Verified Green, Unlinked, Action Required	`S01`, `S09`, `S27`
Clinical Macro / Quick-Order Chip	Clinical Charting	URTI, Diabetes, Hypertension sets	Default, Hover, Selected (Auto-populates Rx)	`S11`
Emergency Takeover Banner	Patient Safety	Red Alert Modal, Persistent Sticky Bar	Active Warning, Click-to-Call 112/911 Triggered	`S01`, `S15`, `S25`

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6. Comprehensive Inferred Data Model (Global Architecture)

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-- Extended Relational Data Model for Global & Indian Telehealth Interoperability

-- 1. Patient Dependent Profiles (India/Global Caregiver Pattern)
CREATE TABLE patient_dependents (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    primary_user_id UUID NOT NULL REFERENCES users(id),
    full_name VARCHAR(255) NOT NULL,
    relationship VARCHAR(32) NOT NULL,
    gender VARCHAR(16) NOT NULL,
    date_of_birth DATE NOT NULL,
    known_allergies TEXT[],
    chronic_conditions TEXT[],
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- 2. Multi-Jurisdiction Licensure (US State Boards, GMC UK, Medical Councils India)
CREATE TABLE clinician_licenses (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    clinician_id UUID NOT NULL REFERENCES users(id),
    jurisdiction_country VARCHAR(4) NOT NULL, -- 'IN', 'US', 'GB', 'FR', 'DE'
    state_or_region VARCHAR(64) NOT NULL,     -- 'Karnataka', 'California', 'NHS England'
    licensing_authority VARCHAR(128) NOT NULL, -- 'Karnataka Medical Council', 'Medical Board of CA'
    license_number VARCHAR(64) NOT NULL,
    license_status VARCHAR(32) DEFAULT 'active',
    valid_until DATE NOT NULL,
    verified_at TIMESTAMPTZ,
    UNIQUE(licensing_authority, license_number)
);

-- 3. Insurance & Benefit Eligibility (US EDI 270/271, France Ameli, Germany GKV)
CREATE TABLE patient_insurances (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    patient_id UUID NOT NULL REFERENCES users(id),
    payer_id VARCHAR(64) NOT NULL,
    payer_name VARCHAR(255) NOT NULL,
    policy_number VARCHAR(128) NOT NULL,
    group_number VARCHAR(128),
    national_health_id VARCHAR(64), -- NHS Number, Carte Vitale, eGK, ABHA ID
    coverage_status VARCHAR(32) DEFAULT 'verified',
    copay_amount NUMERIC(10,2) DEFAULT 0.00,
    eligibility_raw_response JSONB,
    verified_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- 4. Multi-Party Consultation Sessions (Amwell Model: Doctor + Patient + Family + Interpreter)
CREATE TABLE consultation_participants (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    session_id UUID NOT NULL REFERENCES teleconsultation_sessions(id),
    participant_type VARCHAR(32) NOT NULL, -- 'patient', 'primary_doctor', 'specialist', 'family', 'interpreter'
    user_id UUID REFERENCES users(id),
    display_name VARCHAR(255) NOT NULL,
    language_spoken VARCHAR(16),
    joined_at TIMESTAMPTZ,
    left_at TIMESTAMPTZ,
    call_duration_seconds INTEGER DEFAULT 0
);

-- 5. Standardized Drug Catalog & Schedules (India MoHFW, US RxNorm, UK dm+d)
CREATE TABLE standardized_drug_catalog (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    jurisdiction_country VARCHAR(4) NOT NULL, -- 'IN', 'US', 'GB', 'FR', 'DE'
    coding_system VARCHAR(32) NOT NULL,       -- 'CDSCO_IN', 'RxNorm_US', 'dmd_UK', 'CIP_FR'
    code_value VARCHAR(64) NOT NULL,
    brand_name VARCHAR(255) NOT NULL,
    generic_name VARCHAR(255) NOT NULL,
    strength VARCHAR(64) NOT NULL,
    dosage_form VARCHAR(32) NOT NULL,
    is_teleconsult_prohibited BOOLEAN DEFAULT FALSE, -- Schedule X (India), DEA Schedule II (US)
```

```
    created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

-- 6. External Health Record Interoperability Mappings (EHR / FHIR)
CREATE TABLE external_ehr_mappings (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    internal_patient_id UUID NOT NULL REFERENCES users(id),
    ehr_system_type VARCHAR(64) NOT NULL, -- 'ABDM_FHIR', 'EPIC_FHIR', 'CERNER_FHIR', 'EMIS_GP_CONNECT'
    external_patient_identifier VARCHAR(255) NOT NULL,
    last_sync_timestamp TIMESTAMPTZ,
    sync_status VARCHAR(32) DEFAULT 'synchronized'
\.
```

7. Out of Scope (What Cannot Be Cloned Without Institutional Authority)

1. **Sovereign Telematics Networks:** eSanjeevani's NIC intranet, UK NHS Spine core routing, and German Gematik hardware Telematikinfrastruktur (Konnektor appliances) require statutory state authority and national health agency accreditation.
2. **National Clearinghouse Switches:** Direct processing of US Surescripts e-prescribing or French SESAM-Vitale claims requires certified clearinghouse intermediary licenses; private platforms must interface via certified healthcare EDI aggregators.
3. **Medical Board Licensure Authority:** Direct real-time verification of doctors across 50 US State Medical Boards or 28 Indian State Medical Councils requires manual credentialing workflows or specialized third-party medical registry verification APIs.
4. **Physical Hospital Real Estate:** Apollo Hospitals' tertiary beds or Amwell's hospital tele-ICU hardware cart fleets represent physical institutional capital that cannot be duplicated in software.

8. Sizing & Implementation Sizing (Global Architecture)

- **Screens:** 28 primary screens (Patient, Clinician, Reception/Triage, Multi-Party Room, Admin, International Insurance).
- **Workflows:** 16 complete workflows (Discovery, Scheduling, Instant Queue, Multi-Party WebRTC, Clinical Charting, e-Rx Routing, EHR Write-Back, Emergency Triage).
- **Entities:** 15 core database entities + 6 auxiliary mapping tables.
- **Complex Engineering Challenges:**
  1. Multi-jurisdiction data residency routing (India DPDP on Azure Central India, EU GDPR on Azure West Europe, US HIPAA on Azure US East).
  2. Multi-party WebRTC SFU media routing with live third-party medical interpreter integration.
  3. HL7 FHIR R4 profile harmonization across ABDM (India), US Core (US), and UK Core (UK).
- **Overall Sizing: XL (Enterprise Large / 24 to 28 weeks to multi-jurisdiction production-ready baseline)**.